Yan DAI

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Education

Massachusetts Institute of Technology Operations Research Center 2024 – Now

PhD student in Operations Research co-advised by Patrick Jaillet and Negin Golrezaei. GPA: **5.0 / 5**.

Tsinghua University Andrew C. Yao Class 2020 – 2024

Bachelor of Engineering in Computer Science and Technology. GPA: **3.99/4.00**.

Research Experiences

Microsoft Research Economics and Computer Science Group (Cambridge, MA) May 2026 – Now

Research intern of Alex Slivkins and Brendan Lucier (also working closely with Nicole Immorlica).

Massachusetts Institute of Technology OR Center (Cambridge, MA) Sep. 2024 – Now

Research assistant of Patrick Jaillet and Negin Golrezaei.

Tsinghua University Andrew C. Yao Class (Beijing, China) Feb. – Jun. 2024

Bachelor's thesis advised by Longbo Huang.

University of Washington School of CSE (Seattle, WA) Jun. – Aug. 2023

Visiting student hosted by Simon S. Du.

Massachusetts Institute of Technology Lab for Info. & Decision (Cambridge, MA) Feb. – May 2023

Visiting student hosted by Suvrit Sra.

University of Southern California CS Department (Remote) Feb. – Dec. 2022

Remote visitor of Haipeng Luo.

Tsinghua University Andrew C. Yao Class (Beijing, China) Jun. – Dec. 2021

Research assistant of Longbo Huang.

Publications

(* stands for equal contribution. Listed reverse chronologically within each category.)

Economics and Computer Science

[15] Yan Dai, Maryam Farboodi, Negin Golrezaei, and Sepehr Shahshahani. "Market Design for AI: Beyond the Copyright Binary."

[14] **[NeurIPS'25]** Yan Dai, Negin Golrezaei, and Patrick Jaillet. "Incentive-Aware Dynamic Resource Allocation under Long-Term Cost Constraints." [Preprint]
1st Place in ACM Student Research Competition (SRC) @ SIGMETRICS'25.

[13] **[COLT'25]** Yan Dai, Moïse Blanchard, and Patrick Jaillet. "Non-Monetary Mechanism Design without Priors: Achieving Efficiency via Adaptive Costly Audits." [Preprint] [Proceeding (Extended Abstract)]

Bandits and Online Learning

[12] Yan Dai, Negin Golrezaei, and Patrick Jaillet. "Policy Regret for Embedding Model Routing: Contextual Bandits with Low-Rank Experts."

[11] **[ACM POMACS & SIGMETRICS'25]** Yan Dai and Longbo Huang. "Adversarial Network Optimization under Bandit Feedback: Maximizing Utility in Non-Stationary Multi-Hop Networks." In *Proceedings of the ACM on Measurement and Analysis of Computing Systems* (ACM POMACS), 8(3):31, 2024. [ACM Journal Page]
Best Paper Award of ACM SIGMETRICS'25.

[10] **[ICLR'25]** Yu Chen*, Jiatai Huang*, Yan Dai*, and Longbo Huang. "uniINF: Best-of-Both-Worlds Algorithm for Parameter-Free Heavy-Tailed MABs." *Spotlight (5%)*. [Preprint] [OpenReview]

[9] **[ICML'23]** Jiatai Huang*, Yan Dai*, and Longbo Huang. "Banker Online Mirror Descent: A Universal Approach for Delayed Online Bandit Learning." [Preprint] [Proceeding]

[8] **[ICLR'23]** Yan Dai, Ruosong Wang, and Simon S. Du. "Variance-Aware Sparse Linear Bandits." [Preprint] [OpenReview]

[7] **[ICML'22]** Jiatai Huang*, Yan Dai*, and Longbo Huang. "Adaptive Best-of-Both-Worlds Algorithm for Heavy-Tailed Multi-Armed Bandits." [Preprint] [Proceeding]

Reinforcement Learning Theory

[6] Aarush Kulkarni, Khang Nguyen, Ricardo Parada, Kenny Guo, William Chang, and Yan Dai. "Learning Adversarial Continuous MDPs with Bandit Feedback and Unknown Transitions."

[5] **[COLT'24]** Yan Dai, Qiwen Cui, and Simon S. Du. "Refined Sample Complexity for Markov Games with Independent Linear Function Approximation." [Preprint] [Proceeding (Extended Abstract)]

[4] **[ICML'23]** Yan Dai, Haipeng Luo, Chen-Yu Wei, and Julian Zimmert. "Refined Regret for Adversarial MDPs with Linear Function Approximation." [Preprint] [Proceeding]

[3] **[NeurIPS'22]** Yan Dai, Haipeng Luo, and Liyu Chen. "Follow-the-Perturbed-Leader for Adversarial Markov Decision Processes with Bandit Feedback." [Preprint] [OpenReview] [Proceeding]

Deep Learning Theory

[2] **[ICML'24]** Kwangjun Ahn, Zhiyu Zhang, Yunbum Kook, and Yan Dai. "Understanding Adam Optimizer via Online Learning of Updates: Adam is FTRL in Disguise." [Preprint] [Proceeding]

[1] **[NeurIPS'23]** Yan Dai*, Kwangjun Ahn*, and Suvrit Sra. "Curious Role of Normalization in Sharpness-Aware Minimization." [Preprint] [OpenReview] [Proceeding]

Research Awards and Scholarships

Research Awards

- Best Paper Award *(1 paper per year)* of ACM SIGMETRICS'25. — Jun. 2025
- 1st Place in ACM Student Research Competition (SRC) @ SIGMETRICS'25. — Jun. 2025

Fellowship and Scholarship

- Runner-Up of Two Sigma PhD Fellowship *(2 winners and 2 runners-up)*. — Feb. 2026
- Outstanding Graduate named by Beijing *(5%)*, by Tsinghua *(2%)*, and by Yao Class *(10%)*. — Jun. 2024
- Presidential Scholarship *(top scholarship for Tsinghua undergrads; 10/16,000+)*, Tsinghua. — Nov. 2023
- Andrew C. Yao Award – Gold Medal *(top scholarship for seniors; 1 student)*, Yao Class. — Sep. 2023

Competitive Programming

- Gold Medal *(1st place with perfect score)*, Asia-Pacific Informatics Olympiad (APIO). — May 2019
- Gold Medal *(5th place)*, Chinese National Olympiad in Informatics (NOI). — Jul. 2018
- Gold Medal, Asia-Pacific Informatics Olympiad (APIO). — May 2018
- Gold Medal, IOI China Team Selection Competition (CTSC). — May 2018
- Gold Medal, Chinese Olympiad in Informatics Winter Camp (WC). — Feb. 2018

Miscellaneous

- 1st Place, MIT Intramurals League on Water Polo. — Apr. 2023
- 2nd Place, National Collegiate Water Polo Championships. — Jun. 2021
- 1st Place, Tsinghua Swimming Competition on Men's 100m Butterfly. — Nov. 2021
- 3rd Place, Tsinghua Swimming Competition on Men's 200m Medley. — Nov. 2021

Professional Services

Journal Reviewing. Journal of Machine Learning Research (JMLR), IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI), Machine Learning (ML); IEEE/ACM Transactions on Networking (TON), Performance Evaluation (PEVA), IEEE Internet of Things Journal (IoT-J).

Conference Reviewing. COLT 2025; ICML 2026 (Gold reviewer) / 2025 / 2023; NeurIPS 2026 / 2025 / 2024 / 2023 / 2022; ICLR 2025 / 2024; AISTATS 2026 / 2025 / 2023 / 2022; AAAI 2026; ALT 2023.